

A peer review of:

Bayesian Inference and Reinforcement Learning for Autonomous pH Adjustment in Diverse Chemical Systems

Overview

The authors present a policy-based pH control system for batch chemistry. Their approach is interesting in how it generates a powerful but slow controller which can then be distilled into a lightweight but still accurate model—as evinced by demonstrations in three diverse, pH-adjustment experimental campaigns. Given that the use of Bayesian inference and reinforcement learning is well-precedented in the context of process control, the clarity of the manuscript's impact would be improved by clarifying the unique contributions of this clever approach. Please see the below for some comments to improve the clarity of the work.

Major Comments

1. The claim that “this work establishes a general strategy for transforming pH adjustment from empirical trial-and-error into a structured and learnable chemical decision process” feels tenuous as the manuscript currently stands. While the Bayesian policy can learn from theory/simulation, the imitation and RL policies are fundamentally based in empirical trial-and-error.
 - a. Other than using cleverly simulated data and Bayesian inference to create a template for an RL algorithm, how does this work stand out from other, prior RL-based pH adjustment approaches? For example, Hermansson and Syafie have, in follow-up work to reference 11, demonstrated joint, continuous pH–liquid level control using Bayesian and RL approaches (cf. “Bayesian-ANN controller for pH-control”, *AJPC*, vol. 2, no. 1, pp. 1–6, Dec. 2023 and “Model-free learning control of neutralization processes using reinforcement learning”, *Eng Appl of Artif Intell*, vol. 20, issue 6, pp. 767–782, Sept. 2007)
 - b. Furthermore, in the introduction, the lack of interpretability is named as a key drawback of RL approaches. How does this approach provide interpretability in practice? Are ~8 experiments sufficient to determine how many species are present in a buffer and their pKa values?
 - c. What advantages does the Bayesian policy provide compared to training a neural network directly on simulated data (in theory or in practice)?
2. In the introduction to this manuscript, the study appears to be purely theoretical. The authors may wish to clarify and highlight that physical experiments were performed. This would help reflect the manuscript's impact.
3. The authors highlight the computational costs of the Bayesian Inference approach; however, without a comparison (even if qualitative), it is difficult for the reader to appreciate the necessity or contribution of the distilled reinforcement learning policy. How do the prediction times of these approaches compare to the 20-second mixing + sensor analysis times implemented in the experiments?
4. Based on Figure 2, the Human appears to be more consistent and to use the fewest experiments on average. Is this performance coincidence from the small sample size?

- a. Furthermore, the Human protocol appears somewhat PID-/rule-like. (E.g., “if off by > 2pH units, use 2 mL”, “Use roughly half the previous aliquot after crossing the setpoint”, “when close to the setpoint, switch to O(100 uL) aliquots”). Have the authors considered testing an expert-rule baseline as a way to virtually benchmark performance over a wide range of simulated buffer systems?

Figures

5. Figure 1, panel a: What is the black box occluding all but one curve meant to represent?
6. Figure 2:
 - a. The format of the x-axes is not obvious and may confuse a reader (e.g., “what is the significance of steps 5 and 8 that they get labeled?”). After further inspection, it appears to be uniform spacing at 5 steps and the final step is labeled. Perhaps reducing the major step size to 2 or 3 so that a clear pattern can be established will help in interpretation.
 - b. There is an errant line coming off the top left subplot.
 - c. Consistent pH axes scales (within each row) may better convey the amount of overshoot and rate of convergence.
7. To help the reader fully appreciate the experimental systems having overlapping buffering zones and nontrivial pH-adjustment curves, could (theoretical) titration plots be provided for the example buffered systems (e.g., those presented in Figure 2)? These could be SI figures or insets in Figure 2.

Minor Comments

8. There are two uses of “significantly” without quantification (Section 3 and Experiment 2). Please revise the wording or provide a statistical analysis.
 - a. The former could be quantified based on the information provided in Figure 1.
9. Experiment 1. The phrase “The robotic platforms executed acid or base additions in 50 μ L increments” may be ambiguous: cf. The platform dispenses either 50 μ L of acid or base at each step. While this is easily resolved by looking at the figures, it may be helpful to revise the wording to reference “resolution” instead of “increments”.
10. Experiments 3 and 4. Please clarify which policy is used. (I would assume it is the reinforcement learning)
11. For the training and experiments, can the input volume ranges be provided? Based on inspecting the code, it appears that most are bound [100 uL – 10 mL], but it is difficult to ascertain.
12. That the code is a single ~7000-line document may hinder the reproduction and utility of the work.
13. The current GitHub ReadMe is a template file. The readme should explain the project, the general structure (how items are organized), version requirements for Python and any imported modules (this can instead be done in a separate “requirements.txt” file), and general instructions for how to run the code for those interested in replicating the results.
14. In the equations that use commas, adding a space after each comma would improve legibility.

15. In the SI, some symbols do not render in my PDF viewer, check with the Journal to ensure these are printed correctly. In particular, the paragraph before Table S1.